PHASE 1 – BRAINSTORMING AND IDEATION

AI FAQ ASSISTANT API

AI-Powered FAQ and Customer Support Automation

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1. PROJECT INTRODUCTION

The **AI FAQ Assistant API** is an AI-powered backend system designed to simplify frequently asked question management and provide quick, intelligent answers to users.

The idea behind the project is to combine a traditional FAQ management system with **Artificial Intelligence**. Instead of depending completely on manually written answers, the system can use **Google Gemini AI** to generate answers and create FAQ content based on a given topic.

The application is developed as a **RESTful API** using Node.js and Express.js, with MongoDB used for storing user and FAQ information.

The project focuses on creating a simple, secure, and reusable backend that can later be connected to a web or mobile application.

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2. TEAM MEMBERS

The project is developed by a team of five members.

S.No	Team Member	Role
1	BALRAJ I	Team Lead
2	BALAKESAVAN S	Team Member
3	BALAMURUGAN E	Team Member
4	BHARATH H	Team Member
5	BHUVANESWARAN G	Team Member

The team works together during the different stages of the project, including idea development, requirement analysis, design, development, testing, and documentation.

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3. PROJECT BACKGROUND

Frequently Asked Questions are commonly used by organizations to provide answers to repeated questions.

In a traditional FAQ system, users usually search through a collection of predefined questions and answers. As the amount of information increases, maintaining and searching the FAQ database can become difficult.

At the same time, customer support teams often receive similar questions repeatedly. Preparing answers manually for these questions requires additional time and effort.

The team identified an opportunity to combine a structured FAQ system with AI technology.

The proposed idea is to develop a system where users can search existing FAQs while also receiving AI-assisted answers when a suitable FAQ is not immediately available.

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4. PROBLEM IDENTIFICATION

During the brainstorming stage, the following problems were identified:

- Users frequently ask similar questions.
- FAQ information may become difficult to organize as it grows.
- Searching manually through FAQ information can take time.
- Support teams may repeatedly answer the same questions.
- Creating FAQ content manually requires effort.
- Users may ask questions using different wording.
- A simple FAQ database cannot automatically generate new answers.
- FAQ information requires controlled access and secure management.

These problems formed the basis for developing the **AI FAQ Assistant API**.

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5. BRAINSTORMING AND IDEATION

During brainstorming, several ideas were considered for improving the FAQ and customer support process.

Initial Ideas

1. Develop a digital FAQ management system.
2. Add a search facility for FAQ information.
3. Provide secure user authentication.
4. Allow authorized users to create and maintain FAQs.
5. Add an AI-based question-answering feature.
6. Allow AI to generate new FAQ content.
7. Build the system as a REST API so that different applications can use it.

After considering these ideas, the team decided to combine FAQ management and AI assistance into one backend platform.

Final Idea

AI FAQ Assistant API – AI-Powered FAQ and Customer Support Automation

The final concept combines:

FAQ Management + Search + Authentication + Google Gemini AI

This provides a foundation for an intelligent customer-support and knowledge-management system.

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6. PROPOSED SOLUTION

The proposed system is a backend REST API that manages FAQ information and provides AI-assisted responses.

The basic concept is:

User

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REST API

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FAQ Database

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Search / Information Retrieval

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Google Gemini AI

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Generated Response

The system will allow authenticated users to create and manage FAQ information.

Users can also search the available FAQs. When AI assistance is required, a question or topic can be sent to Google Gemini through the backend.

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7. PROJECT OBJECTIVES

The main objectives identified during the brainstorming stage are:

1. To develop an AI-powered FAQ management API.
2. To provide a simple method for managing FAQ information.
3. To provide secure user authentication.
4. To allow users to search FAQ information.
5. To reduce repetitive customer-support activities.
6. To generate AI-based answers using Google Gemini.
7. To generate FAQ question-answer content using AI.
8. To create a reusable backend API that can later support different applications.

The project is therefore focused on combining conventional knowledge management with AI-based assistance.

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8. TARGET USERS

The initial project concept identifies the following users:

User	Main Purpose
General User	Search FAQs and obtain answers
Authenticated User	Access protected FAQ and AI functions
FAQ Creator	Create and maintain FAQ information
Administrator	Future management and monitoring of the system

The detailed permissions and access control for these users will be defined later in **Phase 2 – Requirement Analysis**.

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9. TECHNOLOGY IDEATION

The team selected technologies that are suitable for developing a RESTful backend application.

Technology	Purpose
Node.js	Backend runtime
Express.js	REST API development
MongoDB	Store application data
Mongoose	Manage MongoDB data models
JWT	User authentication
bcrypt	Password protection
Google Gemini AI	AI-generated answers and FAQ content
Postman	API testing
Visual Studio Code	Development environment

The selected technologies provide the basic foundation required for developing the proposed system.

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10. EXPECTED PROJECT OUTCOME

The expected outcome of the project is a working **AI FAQ Assistant API** that provides:

- Secure user registration and login.
- FAQ creation and management.
- FAQ searching.
- Protected API operations.
- AI-generated answers.
- AI-generated FAQ content.
- MongoDB-based data storage.
- A modular backend structure.
- APIs that can later be connected to a web or mobile frontend.

The final system is intended to provide a foundation for faster FAQ management and AI-assisted customer support.

The project will be developed progressively through the remaining VTPT phases:

Phase	**Activity**
Phase 1	Brainstorming and Ideation
Phase 2	Requirement Analysis
Phase 3	Project Design
Phase 4	Project Planning
Phase 5	Project Development
Phase 6	Project Testing
Phase 7	Project Documentation
Phase 8	Project Demonstration

Phase 1 concludes with the selection of the AI FAQ Assistant concept and establishes the basic direction of the project. Detailed requirements, system design, task planning, implementation, and testing will be covered in the subsequent phases.